Algorithmic fairness

Lecture 3 – Enforcing fairness in binary classification

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Master PSND & Actuariat – 2 December, 2025

Group fairness paradigm

Data: $(\underbrace{\text{feature}}_{\boldsymbol{X}}, \underbrace{\text{sensitive attribute}}_{S}, \underbrace{\text{label}}_{Y}) \sim \mathbb{P}$ on $\mathcal{X} \times \mathcal{S} \times \mathcal{Y}$

Predictions: $g : \mathcal{W} \to \mathcal{Y}$

- Fairness through awareness: $\mathcal{W} = \mathcal{X} \times \mathcal{S}$ (disparate treatment)
- Fairness through UNawareness: $\mathcal{W} = \mathcal{X}$ (legal reasons: regulations)

Risk: $g \mapsto \mathcal{R}(g)$

- classification: $\mathcal{R}(g) = \mathbb{P}(Y \neq g(\boldsymbol{W}))$
- regression: $\mathcal{R}(g) = \mathbb{E}(Y - g(\boldsymbol{W}))^2$

Fairness criteria

Independence: $g(\boldsymbol{W}) \perp\!\!\!\perp S$

Separation: $(g(\boldsymbol{W}) \perp\!\!\!\perp S) \mid Y$

Sufficiency: $(Y \perp\!\!\!\perp S) \mid g(\boldsymbol{W})$

Classification through awareness under DP constraint

Binary classification under DP constraint

Notations

- $\mathcal{S} = \{-1, 1\}$, and $\mathcal{Y} = \{0, 1\}$
- $\pi_s = \mathbb{P}(S = s) > 0$, and $\eta^*(\boldsymbol{X}, S) = \mathbb{P}(Y = 1|\boldsymbol{X}, S)$
- Classifier $g \hookrightarrow$ prediction $g(\boldsymbol{X}, S) \in \{0, 1\}$

Problem

- Demographic Parity (DP) constraint

$$\sum_{s \in \mathcal{S}} s\mathbb{P}\left(g(\boldsymbol{X}, S) = 1|S = s\right) = 0$$

- $g^* \in \arg\min_g \{\ \mathbb{P}(g(\boldsymbol{X}, S) \neq Y),\ g \text{ satisfies DP}\}$
- Lagrangian associated to the minimization problem

$$\mathcal{L}(g, \lambda) \;=\; \mathbb{P}\left(g(\boldsymbol{X}, S) \neq Y\right) + \lambda \sum_{s \in \mathcal{S}} s\mathbb{P}(g(\boldsymbol{X}, S) \;=\; 1|S \;=\; s)$$

Mathematical tools

Duality

- weak duality (always holds)

$$\inf_{g} \sup_{\lambda\in\mathbb{R}} \mathcal{L}(g,\lambda) \geq \sup_{\lambda\in\mathbb{R}} \inf_{g} \mathcal{L}(g,\lambda)$$

- strong duality

$$g^* = \inf_{g} \sup_{\lambda\in\mathbb{R}}, \mathcal{L}(g,\lambda) = \sup_{\lambda\in\mathbb{R}} \inf_{g} \mathcal{L}(g,\lambda)$$

Useful property

- $z \mapsto h(z,W)$ convex, $F(z) = \mathbb{E}\left[h(z,W)\right]$
- F convex and $F'(x) = \mathbb{E}\left[\frac{\partial}{\partial x} h(x,W)\right]$

Optimal predictor under DP constraint: *result*

Continuity assumption

- $t \mapsto \mathbb{P}(\eta^*(\boldsymbol{X}, s) \leq t | S = s)$ is continuous

Optimal predictor

- the optimal fair classifier g^* can be characterized as

$$g^*(\boldsymbol{x}, s) = \mathbb{1}_{\left\{\eta^*(\boldsymbol{x},s) \geq \frac{1}{2} + \frac{s\lambda^*}{2\pi_s}\right\}}$$

- λ^* is Lagrange multiplier characterized as

$$\lambda^* \in \arg\min_{\lambda \in \mathbb{R}} \sum_{s \in \mathcal{S}} \mathbb{E}_{\boldsymbol{X}|S=s} \left[\max\left(\pi_s \left(2\eta^*(\boldsymbol{X}, S) - 1\right) - s\lambda\right), 0\right)\right]$$

Optimal predictor: *sketch of the proof (1/2)*

- for each $\lambda \in \mathbb{R}$, consider the Lagrangian $\mathcal{L}(g, \lambda)$ defined as

$$\mathcal{L}(g, \lambda) = \mathbb{P}(g(\boldsymbol{X}, S) \neq Y) + \lambda \sum_{s \in \{-1,1\}} s \mathbb{P}_{\boldsymbol{X}|S=s}\left(g(\boldsymbol{X}, S) = 1\right)$$

- we have that $\mathcal{L}(g, \lambda)$ can be expressed as

$$\mathbb{E}\left[\eta^*(\boldsymbol{X}, S)\right] - \sum_{s \in \{-1,1\}} \mathbb{E}_{\boldsymbol{X}|S=s}\left[\left(\pi_s(2\eta^*(\boldsymbol{X}, S) - 1) - s\lambda\right) g(\boldsymbol{X}, S)\right]$$

- we deduce that $g^*_\lambda \in \arg\min_g \mathcal{L}(g, \lambda)$ is characterized as

$$g^*_\lambda(\boldsymbol{x}, s) = \mathbb{1}_{\{\pi_s(2\eta^*(\boldsymbol{x}, s) - 1) - s\lambda \geq 0\}},$$

and

$$\mathcal{L}(g^*_\lambda, \lambda) = \mathbb{E}\left[\eta^*(\boldsymbol{X}, S)\right] - \sum_{s \in \mathcal{S}} \mathbb{E}_{\boldsymbol{X}|S=s}\left[\max\left(\pi_s(2\eta^*(\boldsymbol{X}, S) - 1) - s\lambda, 0\right)\right]$$

Optimal predictor: *sketch of the proof (2/2)*

- consider $\lambda^* \arg\min_\lambda H(\lambda)$ with

$$H(\lambda) = \sum_{s \in \mathcal{S}} \mathbb{E}_{\boldsymbol{X}|S=s} \left[\max\left(\pi_s (2\eta^*(\boldsymbol{X}, S) - 1) - s\lambda, 0 \right) \right]$$

- observe that $\lambda^* \in \arg\max \mathcal{L}\left(\lambda, g^*_\lambda\right)$
- under continuity assumption, $\lambda \mapsto H(\lambda)$ is differentiable and the first order condition shows that $g^*_{\lambda^*}$ satisfies DP
- therefore, with the weak duality, we obtain that $g^* = g^*_{\lambda^*}$

First estimation method: *post-processing*

a plug-in approach

Objective

- estimate $g^*(\boldsymbol{x}, s) = \mathbb{1}_{\left\{\eta^*(\boldsymbol{x},s) \geq \frac{1}{2} + \frac{s\lambda^*}{2\pi_s}\right\}}$

Plug-in approach

- labeled sample $\mathcal{D}_n = \{(\boldsymbol{X}_i, S_i, Y_i) : \ i = 1 \ldots, n\} \hookrightarrow$ estimate η^*
- unlabeled sample $(\boldsymbol{X}_{n+1}, S_{n+1}), \ldots, (\boldsymbol{X}_{n+N}, S_{n+N})$
- $\{S_{n+1}, \ldots, S_{n+N}\} \hookrightarrow$ estimate π_s by their empirical frequencies
- $\{\boldsymbol{X}_{n+1}, \ldots, \boldsymbol{X}_{n+N}\} \hookrightarrow$ estimate parameter λ^*

Randomization

- fairness guarantee requires continuity assumption
- introduce $\zeta \sim \mathcal{U}_{[0,u]}$ independent of $(\boldsymbol{X}, S)$, $u \to 0$
- $\bar{\eta}(\boldsymbol{X}, S, \zeta) = \hat{\eta}(\boldsymbol{X}, S) + \zeta$

Randomized classifier

Randomized fair classifier

- $(\boldsymbol{X}_{n+1}, \ldots, \boldsymbol{X}_{n+N}) \hookrightarrow (\boldsymbol{X}_1^s, \ldots, \boldsymbol{X}_{N_s}^s)$ i.i.d. from $\boldsymbol{X}|S = s$
- $(\zeta_1^s, \ldots, \zeta_{N_s}^s)$ i.i.d from ζ
- estimator $\hat{\lambda}$

$$\hat{\lambda} \in \arg\min_{\lambda \in \mathbb{R}} \sum_{s \in \mathcal{S}} \frac{1}{N_s} \sum_{i=1}^{N_s} \max\left(\hat{\pi}_s(2\bar{\eta}(\boldsymbol{X}_i^s, s, \zeta_i^s) - 1) - s\lambda, 0\right)$$

- resulting classifier

$$\hat{g}(\boldsymbol{x}, s) = \mathbb{1}_{\left\{\hat{\eta}(\boldsymbol{x},s) \geq \frac{1}{2} + \frac{s\hat{\lambda}}{2\hat{\pi}_s}\right\}}$$

Theoretical guarantees: *fairness guarantee*

Unfairness measure

$$\mathcal{U}(g) = \left| \sum_{s \in \mathcal{S}} s \mathbb{P}\left(g(\boldsymbol{X}, S) = 1 | S = s\right) \right|$$

Distribution free-result

There exists C depending only on π_s such that for any estimator $\hat{\eta}$

$$\mathbb{E}\left[\mathcal{U}(\hat{g})\right] \leq C N^{-1/2}$$

- Distribution-free result: for any $\mathbb{P}$ and any preliminary estimator of η^*

E. Chzhen *et al.*, *Leveraging Labeled and Unlabeled Data for Consistent Fair Binary Classification*, (2019)

Theoretical guarantees: *consistency*

Measure of performance

- $g^* \in \arg\min_g \mathcal{R}_{\lambda^*}(g) = \mathcal{L}(g, \lambda^*)$

$$\mathcal{R}_{\lambda^*}(g) = \mathbb{P}\left(g(\boldsymbol{X}, S) \neq Y\right) + \lambda^* \sum_{s \in \mathcal{S}} s\mathbb{P}(g(\boldsymbol{X}, S) = 1 | S = s)$$

Theorem

Under continuity assumption

$$\mathbb{E}\left[\mathcal{R}_{\lambda^*}(\hat{g}) - \mathcal{R}_{\lambda^*}(g^*)\right] \lesssim \mathbb{E}\left[|\hat{\eta}(\boldsymbol{X}, S) - \eta^*(\boldsymbol{X}, S)|\right] + u + N^{-1/2}$$

- assume that $\hat{\eta}$ are consistent and $u \to 0$
  $\hookrightarrow$ $\hat{g}$ is consistent

Second estimation method: *in-processing*

Observations

- only one *labeled* sample $(\boldsymbol{X}_i, S_i, Y_i), i = 1, \ldots, n$
- $(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n) \to (\boldsymbol{X}_1^s, \ldots, \boldsymbol{X}_{n_s}^s)$ *i.i.d.* from $\boldsymbol{X}|S = s$

Fair E.R.M.

- let $\mathcal{G}$ a class of classifier
- empirical unfairness constraint, for $\varepsilon > 0$, $g \in \mathcal{G}$

$$\hat{\mathcal{U}}(g) = \left| \sum_{s \in \mathcal{S}} s \frac{1}{n_s} \sum_{i=1}^{n_s} g(\boldsymbol{X}_i^s, s) \right| \leq \varepsilon$$

- empirical risk $\hat{R}(g) = \dfrac{1}{n} \sum_{i=1}^{n} \mathbb{1}_{\{g(\boldsymbol{X}_i, S_i) \neq Y_i\}}$
- empirical risk minimizer

$$\hat{g} \in \arg\min_g \{\hat{R}(g),\ \hat{\mathcal{U}}(g) \leq \hat{\varepsilon}\}$$

In-Processing approach: *properties*

Assumptions

- $\mathcal{G}$ with finite VC dimension $V(\mathcal{G})$, $g^* \in \mathcal{G}$
- classical result

$$\mathbb{E}\left[\sup_{g\in\mathcal{G}}\left|\frac{1}{n}\sum_{i=1}^{n} g(\boldsymbol{X}_i) - \mathbb{E}\left[g(\boldsymbol{X}_i)\right]\right|\right] \leq C\sqrt{\frac{\log(n)}{n}}$$

Theoretical guarantees

- let $\hat{\varepsilon} \propto \sum_{s\in\mathcal{S}} \sqrt{\frac{\log(n)}{n_s}}$
- unfairness $\mathbb{E}\left[\mathcal{U}(\hat{g})\right] \leq C\sqrt{\frac{\log(n)}{n}}$
- risk bound $\mathbb{E}\left[R_{\lambda^*}(\hat{g}) - R_{\lambda^*}(g^*)\right] \leq C\sqrt{\frac{\log(n)}{n}}$

In-Processing approaches: *literature*

Convexification

- convex surrogate of both risk and fairness constraint
- E.R.M. with convex loss

M. Donini *et* al., *Decoupled Classifiers for Group-Fair and Efficient Machine Learning*, (2018)

Randomized classifier

- define a set of distributions over the class of classifier
- sample according to a given distribution μ
- randomized classifier
- E.R.M. with randomized classifiers

A. Agarwal *et* al., *A Reductions Approach to Fair Classification*, (2018)

Regression through awareness under DP constraint

Regression under DP constraint

Fair regression problem

- observation $(\boldsymbol{X}, S, Y)$, $Y \in \mathbb{R}$
- $Y = \eta^*(\boldsymbol{X}, S) + \varepsilon$ with $\mathbb{E}\left[\varepsilon | \boldsymbol{X}, S\right] = 0$
- prediction rule: $f : \mathbb{R}^d \times \mathcal{S} \to \mathbb{R}$
- risk $R(f) = \mathbb{E}\left[(Y - f((\boldsymbol{X}, S))^2\right]$
- *exact* DP constraint

$$\sup_{t \in \mathbb{R}} |\mathbb{P}(f(\boldsymbol{X}, S) \leq t | S = 1) - \mathbb{P}(f(\boldsymbol{X}, S) \leq t | S = -1)| = 0$$

- *optimal fair* predictor f^* defined as

$$f^* \in \arg\min_f \{R(f), f \text{ satisfies DP}\}$$

First approach: *discretization*

Discretization

- assume that $|Y| \leq 1$
- consider a grid $\mathcal{G}_L = \left\{\frac{l}{L},\ l = -L, \dots, L\right\}$, $L > 0$
- discretized predictor $f_L(x, s) \in \mathcal{G}_L$

DP constraint for discretized predictor

- f_L^* statisfies DP *iff*

$$\max_{l \in \{-L, \dots, L\}} \sum_{s \in \mathcal{S}} s \mathbb{P}_{\boldsymbol{X} | S = s}\left(f_L(\boldsymbol{X}, S) = \frac{l}{L}\right) = 0$$

- $f_L^* \arg\min_{f_L} \{R(f_L),\ \ f_L \ \text{satisfies} \ \text{DP}\}$

Approximation property

- we have

$$R(f_L^*) \leq R(f^*) + 2\frac{\sqrt{\mathrm{Var}(Y)}}{L} + \frac{1}{L^2}$$

- proposal : estimate f_L^* rather than f^*

Optimal discretized fair predictor

Continuity assumption

- $t \mapsto \mathbb{P}(\eta^*(\boldsymbol{X}, s) \leq t | S = s)$ is continuous

Optimal predictor

- f^*_L can be characterized as

$$f^*_L \in \arg\min_l \pi_s \left(\eta^*(\boldsymbol{X}, S) - \frac{l}{L}\right) - s\lambda^*_l,$$

with $\lambda^* = (\lambda^*_{-L}, \dots \lambda^*_L)$

$$\lambda^* \in \arg\min_{\lambda \in \mathbb{R}^{2L+1}} \sum_{s \in \mathcal{S}} \mathbb{E}_{\boldsymbol{X}|S=s} \max_l \left(s\lambda - \pi_s\left(\eta^*(\boldsymbol{X}, s) - \frac{l}{L}\right)\right)$$

Estimation

- similar to the post-processing procedure in classification

E. Chzhen et al., *Fair Regression via Plug-in Estimator and Recalibration With Statistical Guarantees*, (2020)

Second approach: *optimal transport*

Wasserstein distance

- let μ, ν two probability distribution on $\mathbb{R}$
- Wasserstein-2 distance

$$\mathcal{W}_2^2\left(\mu, \nu\right) = \inf_{\gamma \in \Gamma_{\mu,\nu}} \int \left|x - y\right|^2 d\gamma(x, y)$$

s.t. $\forall \gamma \in \Gamma(\mu, \nu)$, $\gamma(A \times \mathbb{R}) = \mu(A)$, and $\gamma(\mathbb{R} \times B) = \nu(B)$

Useful characterizations

- if $\boldsymbol{X}$ admits a density ν, there exists a mapping T such that

$$\mathcal{W}_2^2\left(\mu, \nu\right) = \mathbb{E}\left[\left(\boldsymbol{X} - T(\boldsymbol{X})\right)^2\right]$$

with $T = F_\mu^{-1} \circ F_\nu$

Optimal prediction under DP

Optimal fair: $f^* \in \arg\min_f \left\{\mathbb{E}(Y - f(\boldsymbol{X}, S))^2 \,:\, f(\boldsymbol{X}, S) \perp\!\!\!\perp S\right\}$

Bayes optimal: $\eta^* \in \arg\min_h \mathbb{E}(Y - h(\boldsymbol{X}, S))^2$

Question: is there a link between f^* and η^*?

- η is (usually) unfair

Optimal prediction under DP

Optimal fair: $f^* \in \arg\min_f \left\{\mathbb{E}(Y - f(\boldsymbol{X}, S))^2 \;:\; f(\boldsymbol{X}, S) \perp\!\!\!\perp S\right\}$

Bayes optimal: $\eta^* \in \arg\min_h \mathbb{E}(Y - h(\boldsymbol{X}, S))^2$

Question: is there a link between f^* and η^*?

η is (usually) unfair

Optimal fair predictor

Let $\pi_s = \mathbb{P}(S = s)$ and $F_{\eta^*|S=s}(t) = \mathbb{P}(\eta^*(\boldsymbol{X}, S) \leq t | S = s)$.
Assume $\nu_{\eta^*|s} := \mathrm{Law}(\eta^*(\boldsymbol{X}, S) | S = s) \in \mathcal{P}_2(\mathbb{R})$ has a density. Then

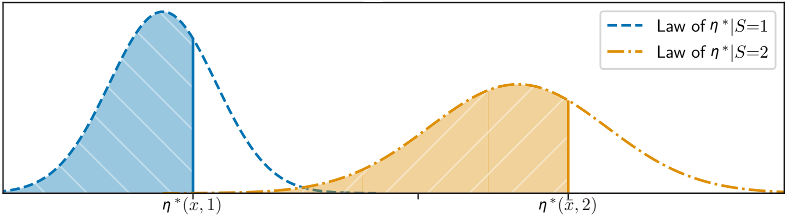
- $$\min_{f \text{ is fair}} \mathbb{E}(\eta^*(\boldsymbol{X}, S) - f(\boldsymbol{X}, S))^2 = \underbrace{\min_{\nu} \sum_{s \in \mathcal{S}} \pi_s \mathcal{W}_2^2(\nu_{\eta^*|s}, \nu)}_{\text{Wasserstein barycenter problem}}$$
- $$f^*(\boldsymbol{x}, s) = \left(\sum_{s' \in \mathcal{S}} \pi_{s'} F^{-1}_{\eta^*|S=s'}\right) \circ F_{\eta^*|S=s} \circ \eta^*(\boldsymbol{x}, s)$$

E. Chzhen et al., *Fair Regression with Wasserstein Barycenters*, (2020)

Interpretation for $\mathcal{S} = \{1, 2\}$

$$\textbf{Fair optimal:} \quad f^*(\boldsymbol{x}, 1) = \pi_1 \eta^*(\boldsymbol{x}, 1) + \pi_2 \underbrace{F^{-1}_{\eta^*|S=2} \circ F_{\eta^*|S=1} \circ \eta^*(\boldsymbol{x}, 1)}_{t^*(\boldsymbol{x},1)}$$

Fair optimal prediction f^* with $\pi_1 = 2/5$ and $\pi_2 = 3/5$



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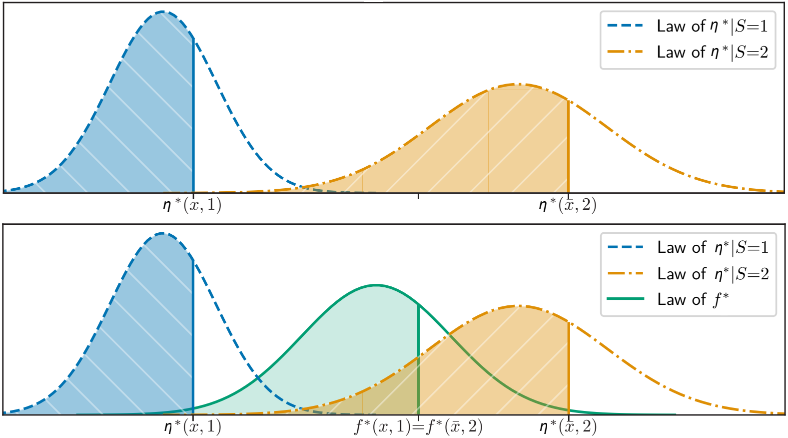

Data-driven procedure

Plug-in procedure

- labeled sample $\mathcal{D}_n \hookrightarrow \hat{\eta}$ + randomization $\hookrightarrow \bar{\eta}$
- unlabeled sample $(\boldsymbol{X}_{n+1}, S_{n+1}), \ldots, (\boldsymbol{X}_{n+N}, S_{n+N})$
- $(S_{n+1}, \ldots, S_{n+N}) \hookrightarrow \hat{\pi}_s$
- $\mathcal{U}_{N_s} = \{\boldsymbol{X}_1^s, \ldots, \boldsymbol{X}_{N_s}^s\} = \mathcal{U}_{N_s}^0 \cup \mathcal{U}_{N_s}^1$
- $\mathcal{U}_{N_s}^0 \hookrightarrow \hat{F}_{\bar{\eta}_{|s}}^{-1}, \mathcal{U}_{N_s}^1 \hookrightarrow \hat{F}_{\bar{\eta}_{|s}}$

Resulting estimator

- $\hat{f}(\boldsymbol{x}, s) = \hat{\pi}_s \hat{f}(\boldsymbol{x}, s) + 1 - \hat{\pi}_s \hat{F}_{\hat{\eta}_{|s'}}^{-1}(\hat{F}_{\hat{\eta}_{|s}}(\hat{\eta}(\boldsymbol{x}, s)))$

Theoretical properties

- same guarantees as in classification.

Takeaway message

Fairness

- presented classification and regression under Demographic Parity
- two main approaches: discretization and optimal transport
- provided a post-processing method with statistical guarantees

Other directions

- prediction when the sensitive attribute cannot be used for prediction (legal reasons)
- specify fairness for a given use case: other notions of fairness should also be considered
- relax the fairness constraint which can be more attractive in some cases since fairness induces risk inflation

Toolboxes for enforcing fairness

AI Fairness 360

Fairlearn

Aequitas
Bias & Fairness Audit

- Have also a look at *What if*

Illustration

$$\mathbb{P}(f(X,S) \in \mathcal{C} | S = -1) = \mathbb{P}(f(X,S) \in \mathcal{C} | S = +1) \ \forall \mathcal{C} \subset \mathbb{R}$$

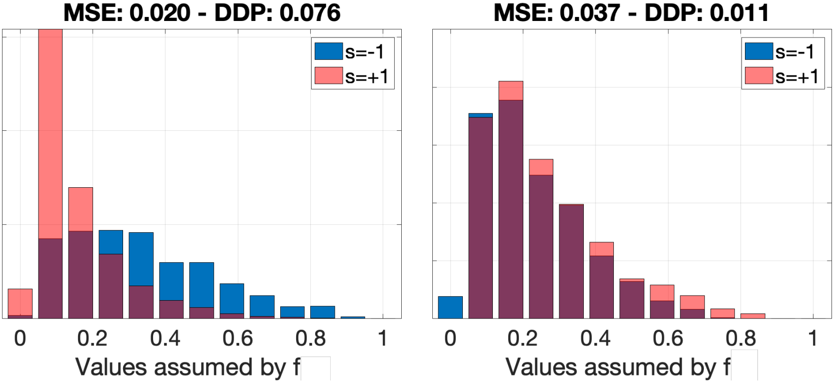

Figure: Algorithm without fairness (left) vs fairness aware (right)

Numerical study

Data. *Communities&Crime* (CRIME); predict the number of violent crimes per 100000 population *w.r.t.* race.
Law School (LAW); School Admissions *w.r.t.* race.
National Longitudinal Survey of Youth (NLSY); predict labor market person's GPA *w.r.t.* race.
Student Performance (STUD); predict students achievement (final grade) *w.r.t.* gender.
University Anonymous (UNIV); predict the average grades at the end of the first semester *w.r.t.* gender.

Methods. *RLS*: Linear methods based on Regularized Least Squares — *KRLS* non-linear models based on Kernel RLS — *RF*: , Random Forests (RF), Random Forests.

	CRIME		LAW		NLSY		STUD		UNIV	
Method	MSE	KS	MSE	KS	MSE	KS	MSE	KS	MSE	KS
RLS	.033±.003	.55±.06	.107±.010	.15±.02	.153±.016	.73±.07	4.77±.49	.50±.05	2.24±.22	.14±.01
RLS+Berk	.037±.004	.16±.02	.121±.013	.10±.01	.189±.019	.49±.05	5.28±.57	.32±.03	2.43±.23	.05±.01
RLS+Oneto	.037±.004	.14±.01	.112±.012	.07±.01	.156±.016	.50±.05	5.02±.54	.23±.02	2.44±.26	.05±.01
RLS+Ours	.041±.004	.12±.01	.141±.014	.02±.01	.203±.019	.09±.01	5.62±.52	.04±.01	2.98±.32	.02±.01
KRLS	.024±.003	.52±.05	.040±.004	.09±.01	.061±.006	.58±.06	3.85±.36	.47±.05	1.43±.15	.10±.01
KRLS+Oneto	.028±.003	.19±.02	.046±.004	.05±.01	.066±.007	.06±.01	4.07±.39	.18±.02	1.46±.13	.04±.01
KRLS+Perez	.033±.003	.25±.02	.048±.005	.04±.01	.065±.007	.08±.01	3.97±.38	.14±.02	1.50±.15	.06±.01
KRLS+Ours	.034±.004	.09±.01	.056±.005	.01±.01	.081±.008	.03±.01	4.46±.43	.03±.01	1.71±.16	.02±.01
RF	.020±.002	.45±.04	.046±.005	.11±.01	.055±.006	.55±.06	3.59±.39	.45±.05	1.31±.13	.10±.01
RF+Raff	.030±.003	.21±.02	.058±.006	.06±.01	.066±.006	.08±.01	4.28±.40	.09±.01	1.38±.12	.02±.01
RF+Agar	.029±.003	.13±.01	.050±.005	.04±.01	.065±.006	.07±.01	3.87±.41	.07±.01	1.40±.13	.02±.01
RF+Ours	.033±.003	.08±.01	.064±.006	.02±.01	.070±.007	.03±.01	4.18±.38	.02±.01	1.49±.14	.01±.01

Note. Our method is very effective in enforcing fairness at the cost of a slight increase in the MSE.